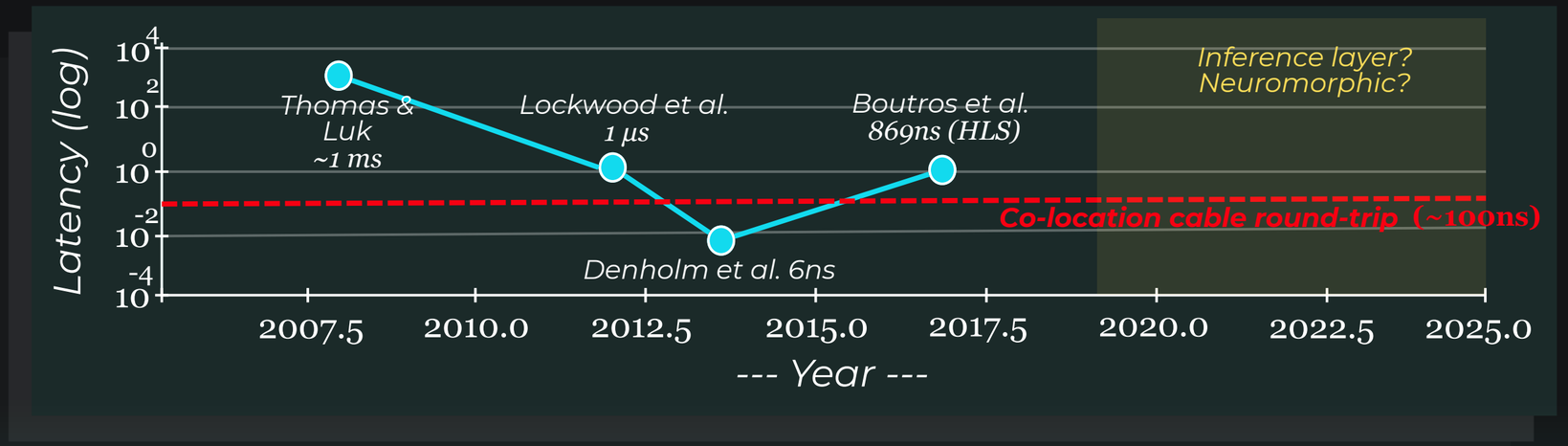
Event-Driven and Many-Threaded Architectures for High-Frequency Trading

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The Problem



Round-trip signal time in co-location cables is already below 100ns

Adding inference logic to an FPGA breaks timing closure

The hardware that gives HFT firms determinism **cannot give them complexity.**

Where does the intelligence go?

Alternative: Event-Driven Architectures

Stimulus-Driven

Computation fires only upon input arrival. No wasted clock cycles

Low Idle Power

Idle regions consume almost no power. No clock toggles logic (asynchronous).

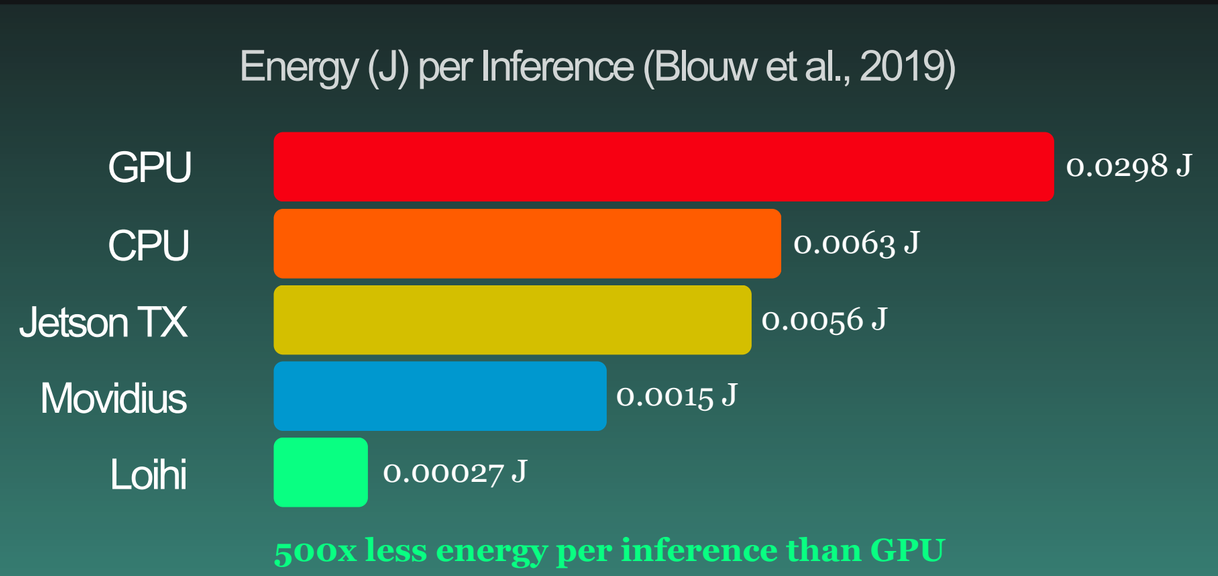
Message Passing

Data moves between adjacent cores, bypassing the memory wall.

Co-location power caps at:

10 kW per rack

Less power at inference = more intelligence in the same rack



SNNs in Financial Trading

Spike-Timing-Dependent Plasticity (STDP)

An unsupervised learning rule where synaptic connections strengthen or weaken based on the relative timing of pre- and post-synaptic spikes. The network learns temporal patterns without being told what to look for.

STDP (unsupervised)

76.8% return

Sharpe 19.71 | Win rate 53.49%

Backpropagation (supervised)

42.5% return

Sharpe 15.91

Independently confirmed

Sharpe ratios above 20 on crude oil and gold futures using the same learning rule [16]

Why this matters

STDP is the learning rule neuromorphic chips run natively. Better signals from the same mechanism that saves power.

Ezinwoke & Rhodes (2025) on AAPL tick data, 19 trading days [10]

What is Missing?

1. No cross-substrate power benchmark on the same financial workload
2. Event-driven financial simulation never run on POETS — **Actionable**
3. No fabricated neuromorphic chip tested on a market data feed
4. SNN trading never validated beyond 19 days of a single equity — **Actionable**
5. Hardware-heterogeneous market microstructure never modelled

The Path Forward

A split pipeline.

FPGA

Deterministic Execution

- Feed parsing
- Arbitration
- Order Submission

Jitter is fatal here

Event-driven messaging

Neuromorphic

Inference Layer

- Pattern recognition
- Signal generation
- Risk assessment

Power efficiency matters here

Fixed 10kW co-location power budget

Every component exists independently. The integration thereof is a natural next step for the intersection of these fields